

Nonlinear-Map Manifold POD Operator Inference

A structure-preserving extension of polynomial-manifold OpInf with KdV and parametric Burgers evidence

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Abstract

We formulate and test a nonlinear-map manifold POD operator-inference model (NM-MPOD) for transport-dominated reduced-order modeling. The construction keeps the modal primary/secondary split used in polynomial-manifold OpInf: the online state is the primary coordinate vector $\mathbf{q}(t; \boldsymbol{\mu}) \in \mathbb{R}^r$, while the secondary POD coordinates are recovered algebraically. The new ingredient is to replace the fixed polynomial secondary map $\Xi \mathbf{g}_p(\mathbf{q})$ by a learned nonlinear map $\mathcal{N}(\mathbf{q}, \boldsymbol{\mu}) \in \mathbb{R}^{\bar{r}}$, represented here by Gaussian-process and neural-network regressors.

The central algebraic point is that, for a quadratic full-order vector field, a nonlinear split decoder changes the correct reduced operator library. Once

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathbf{z}, \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}),$$

the state-quadratic expansion contains not only $\mathbf{q} \otimes \mathbf{q}$ but also $\mathbf{q} \otimes \mathbf{z}$ and $\mathbf{z} \otimes \mathbf{z}$. We therefore distinguish between nonlinear-feature closures, which append a learned secondary feature block, and the state-quadratically complete NM-MPOD library, which includes all three quadratic blocks in $(\mathbf{q}, \mathbf{z})$.

We use the following nomenclature throughout. The *augmented-linear* variant (AL-NM-MPOD) uses a vector field that is affine in the augmented coordinates $[\mathbf{q}^T, \mathbf{z}^T]^T$. The *primary-quadratic* variant (PQ-NM-MPOD) retains the standard primary quadratic block $\mathbf{q}_{\text{quad}}$ and adds $\mathbf{z}$ as a nonlinear feature. The *quadratic-complete* variant (QC-NM-MPOD) is the algebraically complete quadratic library in the augmented coordinates $[\mathbf{q}^T, \mathbf{z}^T]^T$.

The KdV experiments verify the nonlinear-feature and MPOD machinery on a published soliton-style benchmark. The parametric Burgers experiments compare three ANN-NM-MPOD dynamics libraries with $r = 10$ and $\bar{r} = 133$: QC-NM-MPOD, PQ-NM-MPOD, and AL-NM-MPOD. In the retained Burgers suite, all three ANN-NM-MPOD variants reduce the relative rollout errors from approximately 9×10^{-2} for standard OpInf and 8×10^{-2} for polynomial MPOD to the 10^{-2} – 10^{-3} range. The QC variant has the best average error, about 6×10^{-3} , but it uses a much larger operator library: 10495 inferred right-hand-side features versus 199 for AL-NM-MPOD, 254 for PQ-NM-MPOD, 121 for standard quadratic OpInf, and 276 for MPOD with $p = 2$.

1 Overview and contributions

The proposed reduced representation is

$$\mathbf{u}(t; \boldsymbol{\mu}) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}), \quad (1)$$

with $\mathbf{V} \in \mathbb{R}^{N \times r}$, $\bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{r}}$, and

$$\mathcal{N} : \mathbb{R}^{r+n_\mu} \rightarrow \mathbb{R}^{\bar{r}}. \quad (2)$$

Only $\mathbf{q}(t; \boldsymbol{\mu}) \in \mathbb{R}^r$ is evolved by the online ODE. The vector $\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu})$ supplies the coefficients of the secondary POD modes. In the retained Burgers experiments,

$$r = 10, \quad \bar{r} = 133, \quad r + \bar{r} = 143.$$

The three central claims are:

1. The idea is a direct generalization of the MPOD decoder: replace the polynomial secondary-coordinate map $\Xi \mathbf{g}(\mathbf{q})$ by a learned nonlinear map $\mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$ while retaining a modal primary/secondary decomposition.
2. For a quadratic full-order vector field and the modal projection used here, the state-quadratically complete library is not just $\mathbf{q} \otimes \mathbf{q}$. It also includes $\mathbf{q} \otimes \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$ and $\mathcal{N}(\mathbf{q}, \boldsymbol{\mu}) \otimes \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$.
3. The full NM-MPOD library is faithful to the quadratic algebra, but much larger than the standard linear-subspace library. In the Burgers case the feature count is 10495 for QC-NM-MPOD versus 254 for PQ-NM-MPOD, 199 for AL-NM-MPOD, 121 for standard quadratic OpInf, and 276 for induced MPOD with $p = 2$. This makes regularization and validation central rather than optional.

Table 1: Notation used throughout the manuscript. Vectors and matrices are bold; plain symbols denote scalar dimensions or scalar hyperparameters.

Symbol	Meaning
$\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$	Semidiscrete full-order state.
$\boldsymbol{\mu} \in \mathbb{R}^{n_\mu}$	Physical parameter vector; for Burgers, $\boldsymbol{\mu} = (\mu_1, \mu_2)^T$.
$\boldsymbol{\eta}(\boldsymbol{\mu}) \in \mathbb{R}^m$	Chosen parameter feature vector; in Burgers $m = 5$ and $\boldsymbol{\eta} = [\mu_1, \mu_2, \mu_1^2, \mu_1\mu_2, \mu_2^2]^T$.
$\mathbf{V} \in \mathbb{R}^{N \times r}$	Primary POD basis; $\mathbf{q}(t; \boldsymbol{\mu})$ are the evolved coordinates.
$\bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{r}}$	Secondary POD basis; $\bar{\mathbf{q}}(t; \boldsymbol{\mu})$ is reconstructed algebraically.
$\mathbf{g}_p(\mathbf{q})$	Polynomial MPOD decoder features.
$\hat{\mathbf{g}}(\mathbf{q})$	Higher monomials induced in the reduced vector field after substituting the MPOD decoder into a quadratic FOM.
$\mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$	Learned secondary-coordinate map for NM-MPOD.
$\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu})$	OpInf right-hand-side feature vector.
$\mathbf{W}$	Inferred reduced operator matrix satisfying $\dot{\mathbf{q}} \approx \mathbf{W}\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu})$.

2 Relationship to nonlinear-manifold OpInf

Geelen, Balzano, Wright, and Willcox formulate nonlinear modal state approximations of the form

$$\mathbf{s}(t) \approx \mathbf{s}_{\text{ref}} + \mathbf{V}\hat{\mathbf{s}}(t) + \bar{\mathbf{V}}\Xi \mathbf{g}(\hat{\mathbf{s}}(t)), \quad (3)$$

where $\mathbf{V}$ contains primary modal directions, $\bar{\mathbf{V}}$ contains additional orthogonal directions, and $\mathbf{g}$ is typically a low-order polynomial embedding [3]. The present study uses the POD-based construction: $\mathbf{V}$ and $\bar{\mathbf{V}}$ are selected from the POD hierarchy. The corresponding polynomial-manifold baseline is MPOD-OpInf.

The proposed NM-MPOD approximation keeps the modal structure in (3), but replaces the fixed polynomial map:

$$\Xi g(\mathbf{q}) \rightsquigarrow \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}). \quad (4)$$

Thus, the decoder becomes (1). This differs from a generic autoencoder ROM in three important ways:

1. The basis matrices $\mathbf{V}$ and $\bar{\mathbf{V}}$ remain modal and interpretable.
2. The online state dimension is still r , not $r + \bar{r}$.
3. The reduced dynamics can still be learned with an OpInf library whose algebraic terms are chosen from the structure of the governing equations.

3 Benchmark problems and semidiscrete structure

Both benchmarks considered here lead to reduced vector-field libraries governed by quadratic nonlinearities. This is the key structural fact behind the operator choices in the remainder of the study.

3.1 Periodic KdV soliton

The first benchmark is the periodic Korteweg–de Vries equation

$$s_t + \alpha s s_x + \beta s_{xxx} = 0, \quad x \in [-\pi, \pi), \quad t \in [0, 1], \quad (5)$$

with periodic boundary conditions, $\alpha = 4$, $\beta = 1$, and initial condition

$$s(x, 0) = 1 + 24 \operatorname{sech}^2(\sqrt{8}x). \quad (6)$$

The full-order data are generated with a Fourier pseudospectral spatial discretization using $N = 256$ grid points and an exponential time-differencing Runge–Kutta method with time step $\Delta t = 2 \times 10^{-4}$. The interval $t \in [0, 0.2]$ is used for training and $t \in (0.2, 1]$ is treated as prediction. After spatial discretization, the nonlinear term ss_x is quadratic in the state, so the semidiscrete dynamics have the schematic form

$$\dot{\mathbf{u}} = \mathbf{A}\mathbf{u} + \mathbf{H}(\mathbf{u} \otimes \mathbf{u}). \quad (7)$$

3.2 Parametric inviscid Burgers equation

The second benchmark is a parametric one-dimensional inviscid Burgers problem on $x \in [0, 100]$,

$$u_t + \left(\frac{u^2}{2} \right)_x = 0.02 \exp(\mu_2 x), \quad u(x, 0) = 1, \quad (8)$$

with a left inflow state controlled by μ_1 , equivalently a left boundary flux $f(u(0, t)) = \mu_1^2/2$. The parameter vector is

$$\boldsymbol{\mu} = (\mu_1, \mu_2)^T, \quad \mu_1 \in [4.25, 5.50], \quad \mu_2 \in [0.015, 0.030]. \quad (9)$$

The full-order model is a first-order finite-volume discretization with $N = 5000$ cells and backward Euler time integration with $\Delta t = 0.05$ over 500 time steps. For a cell-centered state $\mathbf{u}^n$, one backward Euler step solves a nonlinear residual of the form

$$R_i(\mathbf{u}^{n+1}; \mathbf{u}^n, \boldsymbol{\mu}) = u_i^{n+1} - u_i^n + \frac{\Delta t}{\Delta x_i} (f(u_i^{n+1}) - f(u_{i-1}^{n+1})) - \Delta t 0.02 \exp(\mu_2 x_i) = 0, \quad (10)$$

with $f(u) = u^2/2$ and $f(u_0^{n+1}) = \mu_1^2/2$ at the inflow boundary. The nonlinear flux is quadratic in the state and the source/boundary terms encode the parameter dependence. The nonintrusive OpInf models represent parameter dependence through

$$\boldsymbol{\eta}(\boldsymbol{\mu}) = [\mu_1 \quad \mu_2 \quad \mu_1^2 \quad \mu_1\mu_2 \quad \mu_2^2]^T. \quad (11)$$

3.3 Quadratic semidiscrete template

Suppressing parameter dependence, both examples may be interpreted through the quadratic semidiscrete template

$$\dot{\mathbf{u}} = \mathbf{c} + \mathbf{A}\mathbf{u} + \mathbf{H}(\mathbf{u} \otimes \mathbf{u}), \quad (12)$$

with $\mathbf{u}(t) \in \mathbb{R}^N$. The OpInf models are nonintrusive: the matrices $\mathbf{c}$, $\mathbf{A}$, and $\mathbf{H}$ are not assembled. Their algebraic structure is used only to choose reduced regression libraries. For example, the standard linear-subspace Burgers baseline uses

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \dot{\mathbf{q}} = \mathbf{W}\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu}), \quad (13)$$

with

$$\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \end{bmatrix}. \quad (14)$$

Here $\mathbf{q}_{\text{quad}}$ denotes the vector of unique products $q_i q_j$ for $i \leq j$.

4 Projected algebra: linear POD, MPOD, and NM-MPOD

This section spells out the algebra that motivates the operator libraries. The formulas are written as if the semidiscrete operators $\mathbf{c}$, $\mathbf{A}$, $\mathbf{H}$ were known, but the actual computation remains nonintrusive: OpInf only uses the resulting term structure to choose a regression library.

For readability, parameter dependence is suppressed in the derivation. The analysis therefore begins from

$$\dot{\mathbf{u}} = \mathbf{c} + \mathbf{A}\mathbf{u} + \mathbf{H}(\mathbf{u} \otimes \mathbf{u}), \quad (15)$$

with $\mathbf{u}(t) \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^{N \times N}$, and $\mathbf{H} \in \mathbb{R}^{N \times N^2}$. Parameter features such as $\boldsymbol{\eta}(\boldsymbol{\mu})$ are then added to the inferred library for the parametric Burgers problem.

4.1 Standard linear-subspace OpInf

For the standard POD approximation,

$$\mathbf{u}(t) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t), \quad \mathbf{V} \in \mathbb{R}^{N \times r}, \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}. \quad (16)$$

The time derivative is

$$\dot{\mathbf{u}} = \mathbf{V}\dot{\mathbf{q}}. \quad (17)$$

Projecting (15) with $\mathbf{V}^T$ gives

$$\dot{\mathbf{q}} = \mathbf{V}^T \mathbf{c} + \mathbf{V}^T \mathbf{A}(\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}) + \mathbf{V}^T \mathbf{H}((\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}) \otimes (\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q})). \quad (18)$$

Now expand the Kronecker product:

$$(\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}) \otimes (\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}) = \mathbf{u}_{\text{ref}} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} \otimes \mathbf{V}\mathbf{q}. \quad (19)$$

Therefore the reduced equation can be grouped as

$$\dot{\mathbf{q}} = \hat{\mathbf{c}} + \hat{\mathbf{A}}\mathbf{q} + \hat{\mathbf{H}}(\mathbf{q} \otimes \mathbf{q}), \quad (20)$$

where the intrusive expressions would be

$$\hat{\mathbf{c}} = \mathbf{V}^T (\mathbf{c} + \mathbf{A}\mathbf{u}_{\text{ref}} + \mathbf{H}(\mathbf{u}_{\text{ref}} \otimes \mathbf{u}_{\text{ref}})), \quad (21)$$

$$\hat{\mathbf{A}}\mathbf{q} = \mathbf{V}^T \mathbf{A}\mathbf{V}\mathbf{q} + \mathbf{V}^T \mathbf{H}(\mathbf{u}_{\text{ref}} \otimes \mathbf{V}\mathbf{q}) + \mathbf{V}^T \mathbf{H}(\mathbf{V}\mathbf{q} \otimes \mathbf{u}_{\text{ref}}), \quad (22)$$

$$\hat{\mathbf{H}}(\mathbf{q} \otimes \mathbf{q}) = \mathbf{V}^T \mathbf{H}(\mathbf{V}\mathbf{q} \otimes \mathbf{V}\mathbf{q}). \quad (23)$$

The important consequence is simple: a quadratic FOM and a linear decoder produce only constant, linear, and quadratic functions of the primary coordinates $\mathbf{q}$. This is the reason the standard OpInf model uses the library

$$\boldsymbol{\theta}_{\text{lin}}(\mathbf{q}, \boldsymbol{\mu}) = [1 \quad \boldsymbol{\eta}(\boldsymbol{\mu}) \quad \mathbf{q} \quad \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \quad \mathbf{q}_{\text{quad}}]^T.$$

4.2 Polynomial MPOD-OpInf

The MPOD nonlinear manifold used in nonlinear-manifold OpInf has the form

$$\mathbf{u}(t) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t) + \bar{\mathbf{V}}\mathbf{z}_p(\mathbf{q}(t)), \quad \mathbf{z}_p(\mathbf{q}) = \boldsymbol{\Xi}\mathbf{g}_p(\mathbf{q}), \quad (24)$$

where $\mathbf{V} \in \mathbb{R}^{N \times r}$ contains primary POD modes, $\bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{r}}$ contains secondary POD modes, and $\mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}$. The vector $\mathbf{g}_p(\mathbf{q})$ contains polynomial features up to degree p , usually starting at degree two. Thus $\mathbf{z}_p(\mathbf{q})$ is not evolved by an ODE; it is reconstructed algebraically from $\mathbf{q}$.

The left-hand side is where an important Willcox-style simplification enters. By the chain rule,

$$\dot{\mathbf{u}} = \mathbf{V}\dot{\mathbf{q}} + \bar{\mathbf{V}} \frac{\partial \mathbf{z}_p}{\partial \mathbf{q}}(\mathbf{q})\dot{\mathbf{q}}. \quad (25)$$

Projecting with $\mathbf{V}^T$ gives

$$\mathbf{V}^T \dot{\mathbf{u}} = \mathbf{V}^T \mathbf{V}\dot{\mathbf{q}} + \mathbf{V}^T \bar{\mathbf{V}} \frac{\partial \mathbf{z}_p}{\partial \mathbf{q}}(\mathbf{q})\dot{\mathbf{q}} = \dot{\mathbf{q}}. \quad (26)$$

So under modal projection with orthogonal $\mathbf{V}$ and $\bar{\mathbf{V}}$, the nonlinear tangent term does not appear on the left-hand side. If instead one used a tangent-space projection, a state-dependent mass matrix would appear. The present study follows the modal projection structure used in the nonlinear-manifold OpInf construction.

Now define the combined modal coordinate

$$\mathbf{y}(\mathbf{q}) = \begin{bmatrix} \mathbf{q} \\ \mathbf{z}_p(\mathbf{q}) \end{bmatrix}, \quad \mathbf{B} = [\mathbf{V} \quad \bar{\mathbf{V}}], \quad \mathbf{u} = \mathbf{u}_{\text{ref}} + \mathbf{B}\mathbf{y}. \quad (27)$$

Substituting into the projected FOM gives

$$\dot{\mathbf{q}} = \mathbf{V}^T \mathbf{c} + \mathbf{V}^T \mathbf{A}(\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{y}_q + \bar{\mathbf{V}}\mathbf{y}_z) + \mathbf{V}^T \mathbf{H}((\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{y}_q + \bar{\mathbf{V}}\mathbf{y}_z) \otimes (\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{y}_q + \bar{\mathbf{V}}\mathbf{y}_z)), \quad (28)$$

where $\mathbf{y}_q = \mathbf{q}$ and $\mathbf{y}_z = \mathbf{z}_p(\mathbf{q})$. Expanding the quadratic term gives

$$\begin{aligned} \mathbf{u} \otimes \mathbf{u} = & \mathbf{u}_{\text{ref}} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \bar{\mathbf{V}}\mathbf{z}_p + \bar{\mathbf{V}}\mathbf{z}_p \otimes \mathbf{u}_{\text{ref}} \\ & + \mathbf{V}\mathbf{q} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \bar{\mathbf{V}}\mathbf{z}_p + \bar{\mathbf{V}}\mathbf{z}_p \otimes \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathbf{z}_p \otimes \bar{\mathbf{V}}\mathbf{z}_p. \end{aligned} \quad (29)$$

After applying $\mathbf{V}^T \mathbf{A}$ and $\mathbf{V}^T \mathbf{H}$, the projected dynamics have the split form

$$\begin{aligned} \dot{\mathbf{q}} = & \mathbf{c}_0 + \mathbf{A}_q \mathbf{q} + \mathbf{A}_z \mathbf{z}_p(\mathbf{q}) + \mathbf{H}_{qq}(\mathbf{q} \otimes \mathbf{q}) \\ & + \mathbf{H}_{qz}(\mathbf{q} \otimes \mathbf{z}_p(\mathbf{q})) + \mathbf{H}_{zz}(\mathbf{z}_p(\mathbf{q}) \otimes \mathbf{z}_p(\mathbf{q})). \end{aligned} \quad (30)$$

Here $\mathbf{H}_{qz}$ absorbs both cross terms $\mathbf{V}\mathbf{q} \otimes \bar{\mathbf{V}}\mathbf{z}_p$ and $\bar{\mathbf{V}}\mathbf{z}_p \otimes \mathbf{V}\mathbf{q}$; formally this uses a permutation between $\mathbf{z}_p \otimes \mathbf{q}$ and $\mathbf{q} \otimes \mathbf{z}_p$.

Equation (30) is the expanded version. The Willcox polynomial-manifold model often writes the same content after regrouping by polynomial degree:

$$\dot{\mathbf{q}} = \hat{\mathbf{c}} + \hat{\mathbf{A}}\mathbf{q} + \hat{\mathbf{H}}(\mathbf{q} \otimes \mathbf{q}) + \hat{\mathbf{P}}\hat{\mathbf{g}}(\mathbf{q}). \quad (31)$$

The subtle point is that $\hat{\mathbf{g}}(\mathbf{q})$ is not necessarily the same vector as the decoder feature vector $\mathbf{g}_p(\mathbf{q})$. It collects the higher-order monomials induced by the substituted nonlinear manifold and by the quadratic FOM. For example, if $\mathbf{z}_p(\mathbf{q})$ contains polynomial terms of degrees $2, \dots, p$, then:

- $\mathbf{A}_z \mathbf{z}_p(\mathbf{q})$ contributes degrees $2, \dots, p$;
- $\mathbf{q} \otimes \mathbf{z}_p(\mathbf{q})$ contributes degrees $3, \dots, p+1$;
- $\mathbf{z}_p(\mathbf{q}) \otimes \mathbf{z}_p(\mathbf{q})$ contributes degrees $4, \dots, 2p$.

The degree-two parts can be absorbed into $\hat{\mathbf{H}}(\mathbf{q} \otimes \mathbf{q})$, while the remaining degree-three through degree- $2p$ terms are collected in $\hat{\mathbf{P}}\hat{\mathbf{g}}(\mathbf{q})$. This is the origin of the higher-order polynomial operator in MPOD-OpInf.

4.3 Nonlinear-map MPOD

The proposed NM-MPOD decoder keeps the same modal split but replaces the fixed polynomial map by a learned map:

$$\mathbf{u}(t; \boldsymbol{\mu}) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}}\mathbf{z}(t; \boldsymbol{\mu}), \quad \mathbf{z}(t; \boldsymbol{\mu}) = \mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}). \quad (32)$$

The chain-rule calculation is identical:

$$\dot{\mathbf{u}} = \mathbf{V}\dot{\mathbf{q}} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}}{\partial \mathbf{q}}(\mathbf{q}, \boldsymbol{\mu}) \dot{\mathbf{q}}, \quad (33)$$

and modal projection again gives

$$\mathbf{V}^T \dot{\mathbf{u}} = \dot{\mathbf{q}} \quad \text{if } \mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}. \quad (34)$$

Thus the Jacobian of the learned secondary map is not needed in this modal-projection OpInf construction.

Substitution into the quadratic FOM gives exactly the same split algebra as (30), but now $\mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$ is not a known polynomial:

$$\begin{aligned} \dot{\mathbf{q}} = & \mathbf{c}_0 + \mathbf{A}_q \mathbf{q} + \mathbf{A}_z \mathbf{z} + \mathbf{H}_{qq}(\mathbf{q} \otimes \mathbf{q}) \\ & + \mathbf{H}_{qz}(\mathbf{q} \otimes \mathbf{z}) + \mathbf{H}_{zz}(\mathbf{z} \otimes \mathbf{z}), \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}). \end{aligned} \quad (35)$$

Equivalently, the quadratic expansion before grouping is

$$\begin{aligned} \mathbf{u} \otimes \mathbf{u} = & \mathbf{u}_{\text{ref}} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \bar{\mathbf{V}}\mathbf{z} + \bar{\mathbf{V}}\mathbf{z} \otimes \mathbf{u}_{\text{ref}} \\ & + \mathbf{V}\mathbf{q} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \bar{\mathbf{V}}\mathbf{z} + \bar{\mathbf{V}}\mathbf{z} \otimes \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathbf{z} \otimes \bar{\mathbf{V}}\mathbf{z}. \end{aligned} \quad (36)$$

This is why adding only $\mathbf{q} \otimes \mathbf{q}$ is not the complete quadratic expansion once the decoder contains a secondary nonlinear map.

Because $\mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$ is arbitrary, (35) cannot be reduced to a finite polynomial in $\mathbf{q}$ of known degree. The final NM-MPOD model therefore uses an OpInf feature library built from the observable reduced quantities $\mathbf{q}$, $\boldsymbol{\mu}$, and $\mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$:

$$\boldsymbol{\theta}_{\text{NM}} = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \\ \mathbf{q} \otimes \mathbf{z} \\ \mathbf{z}_{\text{quad}} \end{bmatrix}. \quad (37)$$

Here $\mathbf{q}_{\text{quad}}$ and $\mathbf{z}_{\text{quad}}$ denote vectors of unique products $q_i q_j$ and $z_i z_j$ for $i \leq j$. The library (37) is the state-quadratic completion of the quadratic FOM after the split decoder (32).

4.4 Why not always use the full library?

The QC library (37) is the most faithful state-quadratic expansion of (36). However, it is also much larger. If $r = 10$ and $\bar{r} = 133$, then

$$\begin{aligned} \dim(\mathbf{q}_{\text{quad}}) &= \frac{10 \cdot 11}{2} = 55, \\ \dim(\mathbf{q} \otimes \mathbf{z}) &= 10 \cdot 133 = 1330, \\ \dim(\mathbf{z}_{\text{quad}}) &= \frac{133 \cdot 134}{2} = 8911. \end{aligned}$$

The QC ANN-NM-MPOD Burgers library therefore has 10495 scalar features. Because only 4509 time samples are available in the nine training trajectories, the regression is underdetermined unless strongly regularized. The quadratic-complete Burgers fit is therefore solved in an equivalent dual ridge form.

There are also two scope boundaries. First, (37) is a state-quadratic completion, not a complete parametric-operator expansion. A fully expanded parametric quadratic model would also consider parameter interactions with $\mathbf{z}$, $\mathbf{q} \otimes \mathbf{z}$, and $\mathbf{z} \otimes \mathbf{z}$, depending on how parameter dependence enters the semidiscrete operators. These terms are excluded from the present Burgers experiments so that the comparison isolates the state-quadratic completion. Second, a tangent-space Galerkin projection of a nonlinear decoder would involve $\mathbf{J}_{\Gamma}(\mathbf{q}) = \partial \Gamma / \partial \mathbf{q}$ and a $\mathbf{q}$ -dependent mass matrix. The present construction instead follows the modal projection spirit of nonlinear-manifold OpInf: the reduced dynamics are inferred directly in the primary modal coordinates using a structured library. This keeps the method nonintrusive and avoids differentiating the ANN decoder, but it is a modeling choice that should be stated explicitly.

5 Nonintrusive training problems

The previous section explains which terms should appear in the reduced model. This section states the actual nonintrusive learning problems. Let

$$\mathbf{q}_j = \mathbf{V}^T(\mathbf{u}_j - \mathbf{u}_{\text{ref}}) \in \mathbb{R}^r, \quad \bar{\mathbf{q}}_j = \bar{\mathbf{V}}^T(\mathbf{u}_j - \mathbf{u}_{\text{ref}}) \in \mathbb{R}^{\bar{r}}, \quad j = 1, \dots, M,$$

be the primary and secondary modal coordinates of the training snapshots. Define

$$\mathbf{Q} = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_M], \quad \bar{\mathbf{Q}} = [\bar{\mathbf{q}}_1 \ \cdots \ \bar{\mathbf{q}}_M], \quad \dot{\mathbf{Q}} = [\dot{\mathbf{q}}_1 \ \cdots \ \dot{\mathbf{q}}_M].$$

For continuous-time OpInf, $\dot{\mathbf{Q}}$ is estimated from the training trajectories by finite differences. For a feature vector $\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu}) \in \mathbb{R}^F$, define

$$\boldsymbol{\Theta} = [\boldsymbol{\theta}(\mathbf{q}_1, \boldsymbol{\mu}_1) \ \cdots \ \boldsymbol{\theta}(\mathbf{q}_M, \boldsymbol{\mu}_M)] \in \mathbb{R}^{F \times M}.$$

The reduced operator is a matrix $\mathbf{W} \in \mathbb{R}^{r \times F}$ such that

$$\dot{\mathbf{q}} \approx \mathbf{W}\boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu}).$$

5.1 Generic continuous-time OpInf regression

The paper-level regularized OpInf regression is

$$\mathbf{W}^* = \arg \min_{\mathbf{W} \in \mathbb{R}^{r \times F}} \left\| \dot{\mathbf{Q}} - \mathbf{W}\boldsymbol{\Theta} \right\|_F^2 + \|\mathbf{W}\boldsymbol{\Lambda}\|_F^2, \quad (38)$$

where $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_F)$ is a feature-wise Tikhonov regularization matrix. Equivalently, one may solve the augmented least-squares system

$$\min_{\mathbf{W}^T} \left\| \begin{bmatrix} \boldsymbol{\Theta}^T \\ \boldsymbol{\Lambda} \end{bmatrix} \mathbf{W}^T - \begin{bmatrix} \dot{\mathbf{Q}}^T \\ \mathbf{0} \end{bmatrix} \right\|_F^2. \quad (39)$$

In the KdV experiments, separate weights are used for the constant, linear, quadratic, and higher-order blocks. If one uses weights $\lambda_c, \lambda_A, \lambda_H, \lambda_P$, then the objective contains their squares through $\|\mathbf{W}\boldsymbol{\Lambda}\|_F^2$.

For the Burgers experiments, the columns of the row-sample design matrix are centered and scaled before solving. In column notation, this is equivalent to fitting

$$\widetilde{\mathbf{W}}^* = \arg \min_{\widetilde{\mathbf{W}}} \left\| \dot{\mathbf{Q}} - \widetilde{\mathbf{W}}\widetilde{\boldsymbol{\Theta}} \right\|_F^2 + \lambda \left\| \widetilde{\mathbf{W}}_{\text{nonconst}} \right\|_F^2, \quad (40)$$

where $\widetilde{\boldsymbol{\Theta}}$ is the centered/scaled feature matrix. The intercept feature is left unpenalized. When $F > M$, the solver uses the mathematically equivalent dual ridge system.

5.2 Linear-subspace OpInf training

For the nonparametric KdV linear-subspace baseline,

$$\boldsymbol{\theta}_{\text{lin}}(\mathbf{q}) = \begin{bmatrix} 1 \\ \mathbf{q} \\ \mathbf{q}_{\text{quad}} \end{bmatrix}, \quad \mathbf{W}_{\text{lin}} = [\hat{\mathbf{c}} \ \hat{\mathbf{A}} \ \hat{\mathbf{H}}].$$

Thus

$$\mathbf{W}_{\text{lin}}^* = \arg \min_{\mathbf{W}} \left\| \dot{\mathbf{Q}} - \mathbf{W} \begin{bmatrix} 1 \\ \mathbf{Q} \\ \mathbf{Q}_{\text{quad}} \end{bmatrix} \right\|_F^2 + \|\mathbf{W} \mathbf{\Lambda}_{\text{lin}}\|_F^2. \quad (41)$$

For the parametric Burgers continuous baseline used in this study,

$$\boldsymbol{\theta}_{\text{lin}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \end{bmatrix}.$$

This is still called a linear-subspace model because the decoder is linear in $\mathbf{q}$; the reduced vector field is allowed to be quadratic in $\mathbf{q}$ and affine in the chosen parameter features.

5.3 MPOD manifold training and dynamics training

MPOD has two nonintrusive learning stages. First, after choosing $\mathbf{V}, \bar{\mathbf{V}}$ from POD, the secondary coordinate map is fit by ridge regression. With

$$\mathbf{G}_p = [\mathbf{g}_p(\mathbf{q}_1) \quad \cdots \quad \mathbf{g}_p(\mathbf{q}_M)],$$

the polynomial map coefficient matrix is

$$\boldsymbol{\Xi}^* = \arg \min_{\boldsymbol{\Xi} \in \mathbb{R}^{\bar{F} \times n_g}} \|\bar{\mathbf{Q}} - \boldsymbol{\Xi} \mathbf{G}_p\|_F^2 + \gamma \|\boldsymbol{\Xi}\|_F^2. \quad (42)$$

The decoder is then

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \boldsymbol{\Xi}^* \mathbf{g}_p(\mathbf{q}).$$

Second, the reduced dynamics are inferred. In the nonparametric full polynomial-manifold derivation used for KdV, the feature vector is

$$\boldsymbol{\theta}_{\text{MPOD}}(\mathbf{q}) = \begin{bmatrix} 1 \\ \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \hat{\mathbf{g}}(\mathbf{q}) \end{bmatrix},$$

where $\hat{\mathbf{g}}(\mathbf{q})$ contains the higher-order monomials induced by substituting $\boldsymbol{\Xi}^* \mathbf{g}_p(\mathbf{q})$ into the quadratic FOM. The operator fit is

$$\mathbf{W}_{\text{MPOD}}^* = \arg \min_{\mathbf{W}} \left\| \dot{\mathbf{Q}} - \mathbf{W} \begin{bmatrix} 1 \\ \mathbf{Q} \\ \mathbf{Q}_{\text{quad}} \\ \hat{\mathbf{G}} \end{bmatrix} \right\|_F^2 + \|\mathbf{W} \mathbf{\Lambda}_{\text{MPOD}}\|_F^2. \quad (43)$$

This is the regression counterpart of (31). For the parametric Burgers MPOD runs, the library keeps the same parameter block used by the linear-subspace Burgers baseline and then appends the induced manifold monomials:

$$\boldsymbol{\theta}_{\text{MPOD}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \hat{\mathbf{g}}(\mathbf{q}) \end{bmatrix}.$$

Thus the Burgers MPOD model is parametric through $\boldsymbol{\eta}(\boldsymbol{\mu})$ and $\boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q}$; it does not include additional parameter products with the induced higher manifold block $\hat{\mathbf{g}}(\mathbf{q})$.

5.4 NM-MPOD manifold training and dynamics training

NM-MPOD replaces the fixed polynomial secondary map by a learned nonlinear map $\mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$. The map can be trained in several ways.

For an ANN with parameters $\boldsymbol{\psi}$,

$$\boldsymbol{\psi}^* = \arg \min_{\boldsymbol{\psi}} \sum_{j=1}^M \|\bar{\mathbf{q}}_j - \mathcal{N}_{\boldsymbol{\psi}}(\mathbf{q}_j, \boldsymbol{\mu}_j)\|_2^2 + \alpha \|\boldsymbol{\psi}\|_2^2, \quad (44)$$

where α denotes weight decay. The ANN training uses a validation split and early stopping.

For an RBF map, write

$$\mathbf{z}(\mathbf{q}, \boldsymbol{\mu}) \approx \mathbf{C}\boldsymbol{\phi}(\mathbf{q}, \boldsymbol{\mu}), \quad \boldsymbol{\Phi} = [\boldsymbol{\phi}(\mathbf{q}_1, \boldsymbol{\mu}_1) \quad \cdots \quad \boldsymbol{\phi}(\mathbf{q}_M, \boldsymbol{\mu}_M)].$$

Then

$$\mathbf{C}^* = \arg \min_{\mathbf{C}} \|\bar{\mathbf{Q}} - \mathbf{C}\boldsymbol{\Phi}\|_F^2 + \rho \|\mathbf{C}\|_F^2. \quad (45)$$

The RBF experiments grid-search kernel type, shape parameter, and ρ .

For a GPR map, the decoder uses the posterior mean

$$\mathbf{z}(\mathbf{q}, \boldsymbol{\mu}) = \mathbb{E}[\bar{\mathbf{q}} \mid \mathbf{q}, \boldsymbol{\mu}, \mathcal{D}].$$

The GP hyperparameters $\boldsymbol{\chi}$ are selected by marginal likelihood:

$$\boldsymbol{\chi}^* = \arg \min_{\boldsymbol{\chi}} [-\log p(\bar{\mathbf{Q}} \mid \mathbf{Q}, \{\boldsymbol{\mu}_j\}_{j=1}^M, \boldsymbol{\chi})]. \quad (46)$$

The posterior variance can support validation and uncertainty screening, but the reduced dynamics in this study use only the posterior mean.

After the secondary map is fixed, define

$$\mathbf{Z} = [\mathbf{z}(\mathbf{q}_1, \boldsymbol{\mu}_1) \quad \cdots \quad \mathbf{z}(\mathbf{q}_M, \boldsymbol{\mu}_M)].$$

The final full state-quadratic NM-MPOD regression uses

$$\boldsymbol{\Theta}_{\text{NM}} = \begin{bmatrix} \mathbf{1} \\ \mathbf{E}_{\boldsymbol{\eta}} \\ \mathbf{Q} \\ \mathbf{E}_{\boldsymbol{\eta}} \odot \mathbf{Q} \\ \mathbf{Q}_{\text{quad}} \\ \mathbf{Z} \\ \mathbf{Q} \otimes \mathbf{Z} \\ \mathbf{Z}_{\text{quad}} \end{bmatrix}. \quad (47)$$

Here $\mathbf{E}_{\boldsymbol{\eta}}$ denotes the repeated parameter-feature block and $\mathbf{E}_{\boldsymbol{\eta}} \odot \mathbf{Q}$ denotes the samplewise blocks $\boldsymbol{\eta}(\boldsymbol{\mu}_j) \otimes \mathbf{q}_j$. The dynamics fit is

$$\mathbf{W}_{\text{NM}}^* = \arg \min_{\mathbf{W}} \left\| \dot{\bar{\mathbf{Q}}} - \mathbf{W}\boldsymbol{\Theta}_{\text{NM}} \right\|_F^2 + \|\mathbf{W}\boldsymbol{\Lambda}_{\text{NM}}\|_F^2. \quad (48)$$

5.5 Rollout-based model selection

Several numerical studies do not choose the ridge parameter solely from derivative fit. They select regularization and RK4 substepping by training/validation rollout. In idealized notation,

$$(\lambda^*, n_{\text{sub}}^*) = \arg \min_{\lambda \in \mathcal{L}, n_{\text{sub}} \in \mathcal{S}} E_{\text{rollout}}^{\text{val}}(\mathbf{W}^*(\lambda), n_{\text{sub}}). \quad (49)$$

This is not part of the algebraic OpInf derivation; it is a practical stabilization and validation step. It should be reported because it affects the meaning of the final test errors.

5.6 A three-level NM-MPOD operator hierarchy

The nonlinear decoder (32) fixes the reconstruction map, but it does not by itself fix the reduced vector-field library. We therefore distinguish three nested NM-MPOD operator classes. Let

$$\mathbf{y}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} \mathbf{q} \\ \mathbf{z} \end{bmatrix}, \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}).$$

For a nonparametric problem, the hierarchy is

$$\text{AL-NM-MPOD:} \quad \boldsymbol{\theta}_{\text{AL}}(\mathbf{q}) = \begin{bmatrix} 1 \\ \mathbf{q} \\ \mathbf{z} \end{bmatrix}, \quad (50)$$

$$\text{PQ-NM-MPOD:} \quad \boldsymbol{\theta}_{\text{PQ}}(\mathbf{q}) = \begin{bmatrix} 1 \\ \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \end{bmatrix}, \quad (51)$$

$$\text{QC-NM-MPOD:} \quad \boldsymbol{\theta}_{\text{QC}}(\mathbf{q}) = \begin{bmatrix} 1 \\ \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \\ \mathbf{q} \otimes \mathbf{z} \\ \mathbf{z}_{\text{quad}} \end{bmatrix}. \quad (52)$$

Here AL denotes *augmented-linear*: the vector field is affine in the augmented coordinates $\mathbf{y} = [\mathbf{q}^T, \mathbf{z}^T]^T$. PQ denotes *primary-quadratic*: the standard primary quadratic block $\mathbf{q}_{\text{quad}}$ is retained, but the mixed and secondary quadratic blocks are omitted. QC denotes *quadratic-complete*: the reduced library is complete for a quadratic full-order vector field after substituting the split decoder.

For the parametric Burgers example, the same hierarchy is used with the parameter blocks from

(11). Thus

$$\boldsymbol{\theta}_{\text{AL}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{z} \end{bmatrix}, \quad (53)$$

$$\boldsymbol{\theta}_{\text{PQ}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \end{bmatrix}, \quad (54)$$

$$\boldsymbol{\theta}_{\text{QC}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \\ \mathbf{q} \otimes \mathbf{z} \\ \mathbf{z}_{\text{quad}} \end{bmatrix}. \quad (55)$$

This parametric QC library is state-quadratically complete in $[\mathbf{q}^T, \mathbf{z}^T]^T$ for the chosen parameter blocks. It is not claimed to be the most general possible parameter-dependent quadratic library; such a library could additionally include products between $\boldsymbol{\eta}(\boldsymbol{\mu})$ and $\mathbf{z}$ or between $\boldsymbol{\eta}(\boldsymbol{\mu})$ and the mixed quadratic blocks.

Table 2: NM-MPOD operator hierarchy. The three names describe the mathematical content of the reduced vector-field library, not the decoder. All three variants use the same nonlinear split decoder $\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$.

Variant	Nonparametric library	Interpretation
AL-NM-MPOD	$[1, \mathbf{q}, \mathbf{z}]$	Augmented-linear vector field in $[\mathbf{q}^T, \mathbf{z}^T]^T$. Cheapest learned-map model; it tests the benefit of the nonlinear decoder with minimal explicit operator structure.
PQ-NM-MPOD	$[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}]$	Primary-quadratic model. It keeps the standard quadratic block in the evolved coordinates and adds the learned secondary coordinates as nonlinear observables.
QC-NM-MPOD	$[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}, \mathbf{q} \otimes \mathbf{z}, \mathbf{z}_{\text{quad}}]$	Quadratic-complete model in the augmented coordinates. This is the library implied by substituting the nonlinear split decoder into a quadratic semidiscrete FOM under modal projection.

6 Offline and online cost

Let r be the number of evolved primary coordinates, $\bar{r}$ the number of secondary modes, $m = \dim \boldsymbol{\eta}(\boldsymbol{\mu})$ the number of parameter features, and F the number of OpInf right-hand-side features. The online ODE cost is dominated by:

$$\text{cost}_{\text{RHS}} \approx \text{cost}(\mathcal{N}) + \text{cost}(\boldsymbol{\theta}) + O(rF). \quad (56)$$

State reconstruction costs $O(Nr + N\bar{r})$ if full fields are explicitly formed, but this is not required at every internal RHS evaluation.

Table 3: Burgers feature and coefficient counts with $r = 10$, $\bar{r} = 133$, and $m = 5$ parameter features. “Decoder features” counts the dimension of the secondary-coordinate map input basis, not the number of neural network hidden activations.

Model	Decoder features	Learned decoder params	RHS features F	OpInf coeffs rF
Standard quadratic OpInf	0	0	121	1210
AL-NM-MPOD	133 outputs	$\approx 1.44 \times 10^5$	199	1990
PQ-NM-MPOD	133 outputs	$\approx 1.44 \times 10^5$	254	2540
Induced MPOD, $p = 2$	10	1330	276	2760
Induced MPOD, $p = 4$	30	3990	856	8560
Induced MPOD, $p = 5$	40	5320	1281	12810
QC-NM-MPOD	133 outputs	$\approx 1.44 \times 10^5$	10495	104950

The distinction in Table 3 matters. MPOD has a compact, linear-in-parameters decoder $\bar{\mathbf{q}} = \Xi \mathbf{g}_p(\mathbf{q})$, so its learned decoder parameters are $\bar{r} \dim \mathbf{g}_p$. ANN-NM-MPOD has a much richer learned decoder and then a much larger QC OpInf library in $[\mathbf{q}^T, \mathbf{z}^T]^T$. Thus the ANN model does not win by being cheaper; it wins in these tests because the learned secondary map and the QC library represent the transported Burgers profiles much more accurately.

For the ANN used in the Burgers experiment, the map dimensions are

$$12 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 256 \rightarrow 133,$$

where the 12 inputs are 10 primary coordinates plus the two physical parameters. The dense neural-network evaluation uses approximately 1.41×10^5 multiply operations per call. This is more expensive than an elementwise polynomial map, but still small relative to a full-order solve with $N = 5000$ cells and implicit nonlinear iterations.

7 Numerical evidence

7.1 KdV soliton reproduction

The KdV soliton case was used to verify the standard OpInf, MPOD, and learned secondary-map machinery on a benchmark close to the nonlinear-manifold OpInf example studied in [3]. Three GPR-NM-MPOD operator libraries are retained here: QC-NM-MPOD with $[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}, \mathbf{q} \otimes \mathbf{z}, \mathbf{z}_{\text{quad}}]$, PQ-NM-MPOD with $[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}(\mathbf{q})]$, and AL-NM-MPOD with $[1, \mathbf{q}, \mathbf{z}(\mathbf{q})]$. The KdV QC case is therefore the nonparametric GPR analogue of the Burgers QC ANN experiment, with $\mathbf{z} = \mathbf{z}_{\text{GPR}}(\mathbf{q})$. For $r = 5$ and $\bar{r} = 9$, the KdV GPR feature counts are 120 for QC, 30 for PQ, and 15 for AL.

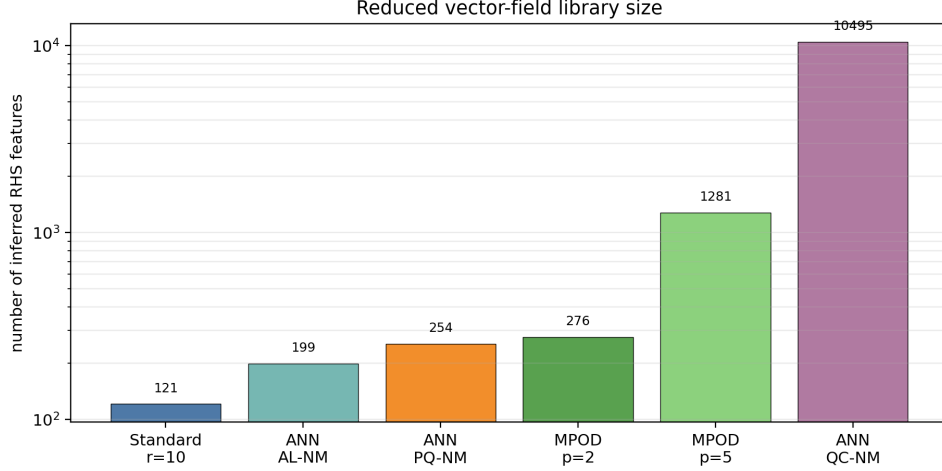


Figure 1: Reduced vector-field feature counts for the Burgers models. The AL- and PQ-NM-MPOD libraries are compact compared with QC-NM-MPOD, but QC-NM-MPOD is the only algebraically complete quadratic library in the augmented variables $[\mathbf{q}^T, \mathbf{z}^T]^T$.

Table 4: Representative KdV soliton errors. The training window is $t \in [0, 0.2]$, and the prediction window is $t \in [0.2, 1]$.

Model	Reconstruction/training error	Full-window error	Prediction-window error
Standard OpInf, $r = 5$	5.14×10^{-1}	5.33×10^{-1}	5.37×10^{-1}
MPOD-OpInf, $r = 5, p = 2$	4.27×10^{-1}	4.44×10^{-1}	4.48×10^{-1}
GPR QC-NM-MPOD, $r = 5, \bar{r} = 9$	8.58×10^{-2}	8.73×10^{-2}	8.77×10^{-2}
GPR PQ-NM-MPOD, $r = 5, \bar{r} = 9$	8.58×10^{-2}	8.73×10^{-2}	8.77×10^{-2}
GPR AL-NM-MPOD, $r = 5, \bar{r} = 9$	8.58×10^{-2}	8.73×10^{-2}	8.77×10^{-2}

7.2 Parametric Burgers test

The primary evidence for the proposed NM-MPOD idea comes from the parametric one-dimensional inviscid Burgers problem. The full-order model uses $N = 5000$ finite-volume cells, backward Euler with $\Delta t = 0.05$, and 501 snapshots per trajectory. Training snapshots are generated on a 3×3 parameter grid:

$$\mu_1 \in \{4.25, 4.875, 5.50\}, \quad \mu_2 \in \{0.015, 0.0225, 0.030\}.$$

The three test points are

$$(4.56, 0.019), \quad (4.75, 0.020), \quad (5.19, 0.026).$$

These three parameter pairs are not used to fit the POD basis, the MPOD or ANN secondary-coordinate maps, or the OpInf operators. They are genuine out-of-sample prediction points in parameter space: each trained ROM is rolled out at the new parameter value and compared against a separately generated HDM trajectory.

8 Interpretation

The retained Burgers results support three conclusions.

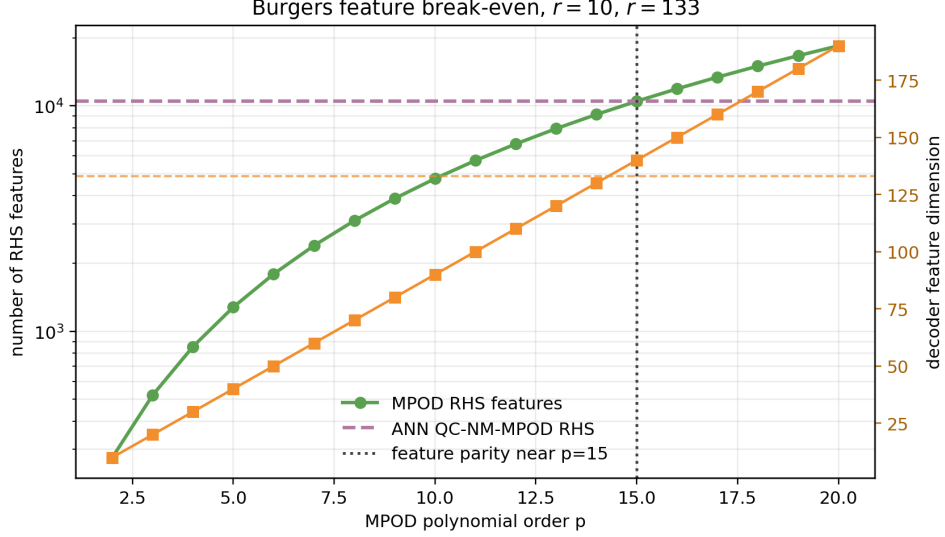


Figure 2: Feature break-even analysis for Burgers with $r = 10$ and $\bar{r} = 133$. Counting only RHS library features, MPOD reaches the QC-NM-MPOD feature count near $p = 15$. Counting decoder features only, MPOD also reaches $\bar{r} = 133$ secondary degrees of freedom near $p = 15$ because $r(p - 1) = 140$. This feature count does not include the ANN evaluation itself; the ANN adds about 1.43×10^5 dense multiply operations per RHS call before the OpInf feature library is evaluated.

1. The nonlinear map is the decisive improvement in this retained Burgers configuration. The standard $r = 10$ linear-subspace OpInf model has errors of order 8.7×10^{-2} to 9.1×10^{-2} . Induced MPOD with $p = 2$ reduces these errors only modestly, to roughly 8.1×10^{-2} to 8.4×10^{-2} . The AL-, PQ-, and QC-NM-MPOD models all reduce the errors by roughly one order of magnitude relative to MPOD. The QC model gives errors of roughly 5.8×10^{-3} to 6.6×10^{-3} , while the AL model gives 6.1×10^{-3} to 1.1×10^{-2} . The MPOD order sweep in Table 6 shows that increasing p from 2 to 5 gives only a small additional improvement. Thus, for this Burgers configuration, the main gain appears to come from the learned secondary map. The mixed/secondary quadratic operator blocks improve the mean test error, but the AL result shows that the decoder itself is already carrying a large part of the gain.

2. Burgers MPOD is now the same kind of MPOD as the KdV MPOD. The Burgers MPOD ODE no longer appends only the decoder powers $g_p(\mathbf{q})$. It uses the induced library $\hat{\mathbf{g}}(\mathbf{q})$ generated by substituting $\Xi \mathbf{g}_p(\mathbf{q})$ into a quadratic full-order vector field. Thus the KdV and Burgers MPOD tests now use the same theoretical construction, with the Burgers library additionally carrying the parameter blocks $\boldsymbol{\eta}(\boldsymbol{\mu})$ and $\boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q}$.

3. QC-NM-MPOD is the algebraically complete quadratic model, while AL-NM-MPOD is the cheapest competitive variant. The ANN QC-NM-MPOD model uses the algebraically complete quadratic library

$$[1, \boldsymbol{\eta}, \mathbf{q}, \boldsymbol{\eta} \otimes \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}, \mathbf{q} \otimes \mathbf{z}, \mathbf{z}_{\text{quad}}], \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}),$$

and gives the lowest average error over the three Burgers test points. The AL variant removes $\mathbf{q}_{\text{quad}}$, $\mathbf{q} \otimes \mathbf{z}$, and $\mathbf{z}_{\text{quad}}$ from the dynamics library, reducing the RHS feature count from 10495 to

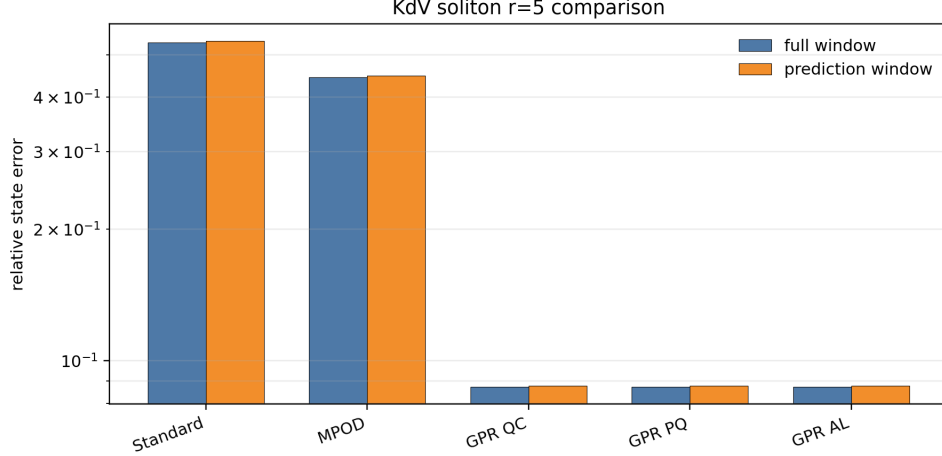


Figure 3: KdV soliton $r = 5$ summary errors for standard OpInf, MPOD, and the three retained nonlinear-feature GPR-NM-MPOD variants.

Table 5: Parametric Burgers relative state errors over all finite rollout snapshots. All models in this table are continuous-time OpInf models and evolve $r = 10$ coordinates. The ANN-NM-MPOD decoder uses $r + \bar{r} = 143$ POD modes with $\bar{r} = 133$ secondary coordinates.

Model	(4.56, 0.019)	(4.75, 0.020)	(5.19, 0.026)
Standard quadratic OpInf, $r = 10$	9.11×10^{-2}	9.11×10^{-2}	8.74×10^{-2}
Induced MPOD, $p = 2$	8.19×10^{-2}	8.39×10^{-2}	8.12×10^{-2}
ANN AL-NM-MPOD	1.12×10^{-2}	8.72×10^{-3}	6.12×10^{-3}
ANN PQ-NM-MPOD	1.19×10^{-2}	1.19×10^{-2}	1.19×10^{-2}
ANN QC-NM-MPOD	5.97×10^{-3}	5.76×10^{-3}	6.56×10^{-3}

199, while remaining competitive and even giving the lowest error at the third test point. This is useful evidence that two effects should be separated in future work: the accuracy of the learned nonlinear decoder and the algebraic completeness of the inferred vector-field library.

9 Scope and proposed next steps

The present results are promising and internally consistent, but they should be read as a preliminary numerical study. A mature journal submission should resolve the following research questions.

1. **Tangent-aware vs. modal projection.** Should NM-MPOD use the same $\mathbf{V}^T$ modal projection implicit in the retained OpInf structure, or should it use a tangent-space projection involving $\mathbf{J}_\Gamma(\mathbf{q})$? The latter is more geometrically faithful but less nonintrusive and more expensive for ANN maps.
2. **Block-regularized QC libraries.** The QC library should be learned with separate penalties for $\mathbf{q}$, $\mathbf{z}$, $\mathbf{q} \otimes \mathbf{q}$, $\mathbf{q} \otimes \mathbf{z}$, and $\mathbf{z} \otimes \mathbf{z}$. A single scalar ridge parameter is too blunt for terms with very different scales and dimensions.
3. **Compressed mixed and secondary quadratic terms.** Instead of using all 1330 $\mathbf{q} \otimes \mathbf{z}$ and

Table 6: Burgers MPOD polynomial-order sensitivity with $r = 10$ and $\bar{r} = 133$. Increasing the polynomial order improves the MPOD rollout errors only modestly while substantially increasing the inferred feature dimension.

MPOD order	RHS features	(4.56, 0.019)	(4.75, 0.020)	(5.19, 0.026)
$p = 2$	276	8.19×10^{-2}	8.39×10^{-2}	8.12×10^{-2}
$p = 4$	856	7.75×10^{-2}	8.30×10^{-2}	8.11×10^{-2}
$p = 5$	1281	7.61×10^{-2}	8.29×10^{-2}	7.97×10^{-2}

8911 $\mathbf{z} \otimes \mathbf{z}$ terms, one can SVD-compress these feature blocks, use randomized sketches, or infer a low-rank operator factorization.

4. **Parameter-complete libraries.** The retained Burgers QC model is state-quadratically complete but not fully parameter-complete. Additional tests should include $\boldsymbol{\eta} \otimes \mathbf{z}$ and possibly $\boldsymbol{\eta} \otimes (\mathbf{q} \otimes \mathbf{z})$ terms if the inferred operator is intended to represent broad parametric dependence.
5. **Decoder uncertainty and validation.** Gaussian-process and RBF versions of $\mathcal{N}$ are useful when data are sparse. GPR uncertainty could be used to detect rollout regions where the ROM leaves the training manifold.
6. **Broader benchmarks.** The present evidence covers KdV and parametric inviscid Burgers. A convincing paper would need additional transport-dominated and nonlinear PDE tests, plus systematic sweeps in $(r, \bar{r})$ and training-set size.

10 Experimental protocol

All numerical results follow the same nonintrusive workflow. First, HDM snapshots are centered by the reference state and compressed by POD. Second, the primary and secondary coordinates are separated according to the prescribed values of r and $\bar{r}$. Third, the secondary-coordinate decoder is fit from training snapshots only. Fourth, the reduced vector field is inferred by regularized continuous-time OpInf using the feature library associated with the chosen model class. Finally, the reduced ODE is initialized from the projected HDM initial condition and integrated over the full prediction horizon without access to the reference trajectory.

For Burgers, the regularization and time-integration hyperparameters are selected using rollout errors on training trajectories; the three parameter points in Table 5 are held out until the final comparison. For KdV, the training interval is $t \in [0, 0.2]$ and the interval $t \in (0.2, 1]$ is used to assess extrapolation in time. These protocols separate model construction from the reported prediction tests and avoid using the plotted test trajectories for operator fitting.

11 Conclusion

The NM-MPOD idea is a natural extension of nonlinear-manifold OpInf: retain the modal primary/secondary split of MPOD, but replace the fixed polynomial secondary-coordinate map by a learned nonlinear map. This keeps the online ODE dimension small while allowing the decoder to represent transport-dominated state geometry more effectively than a low-dimensional linear subspace.

For quadratic PDEs under the modal-projection formulation used here, the quadratic completion is clear: $\mathbf{q} \otimes \mathbf{q}$, $\mathbf{q} \otimes \mathcal{N}$, and $\mathcal{N} \otimes \mathcal{N}$ must be included in the QC-NM-MPOD operator library. The numerical evidence here shows that QC-NM-MPOD combined with an ANN secondary-coordinate map gives the best average continuous-time Burgers error among the retained models, while AL-NM-MPOD is a much cheaper competitive alternative. The next research direction is to learn nonlinear-map manifold decoders and infer compressed, block-regularized, physics-structured reduced operators on top of them.

References

- [1] B. Peherstorfer and K. Willcox. Data-driven operator inference for nonintrusive projection-based model reduction. *Computer Methods in Applied Mechanics and Engineering*, 306:196–215, 2016.
- [2] R. Geelen, S. Wright, and K. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. arXiv:2205.02304, 2022. <https://arxiv.org/abs/2205.02304>.
- [3] R. Geelen, L. Balzano, S. Wright, and K. Willcox. Learning physics-based reduced-order models from data using nonlinear manifolds. arXiv:2308.02802v2, 2024. <https://arxiv.org/abs/2308.02802>.

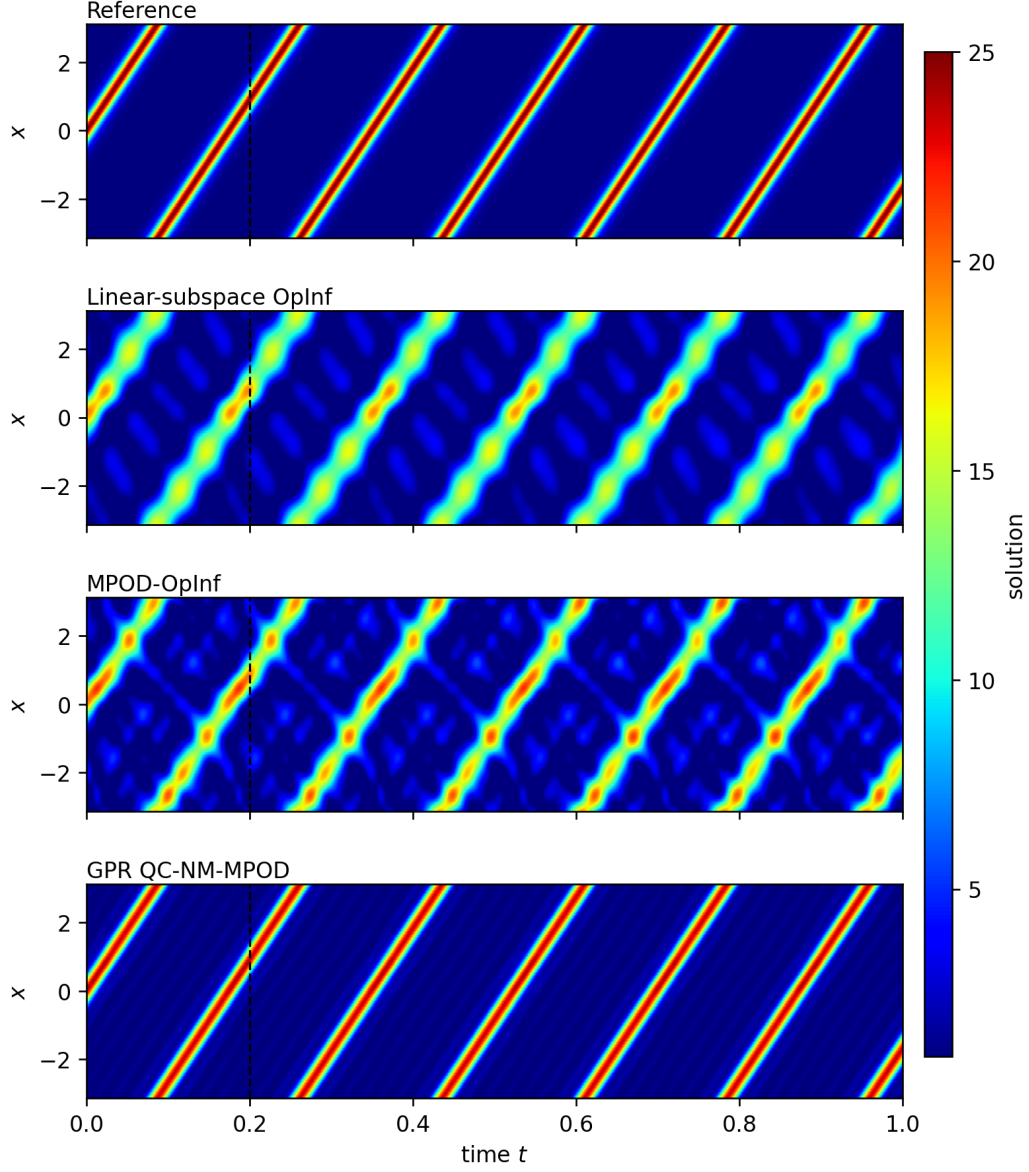


Figure 4: Space-time KdV comparison in a single shared scale: reference, linear-subspace OpInf, MPOD-OpInf, and the GPR QC-NM-MPOD model. The dashed vertical line marks the end of the training window at $t = 0.2$; the interval $t \in (0.2, 1]$ is prediction.

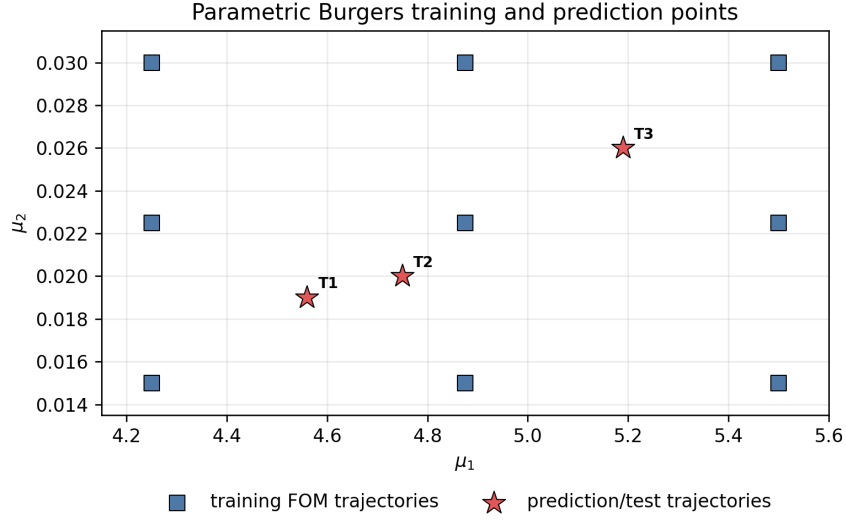


Figure 5: Parametric Burgers training and prediction points. Blue squares are the nine HDM trajectories used to build the POD basis and fit the reduced models. Red stars are the three unseen parameter points used only for final prediction tests.

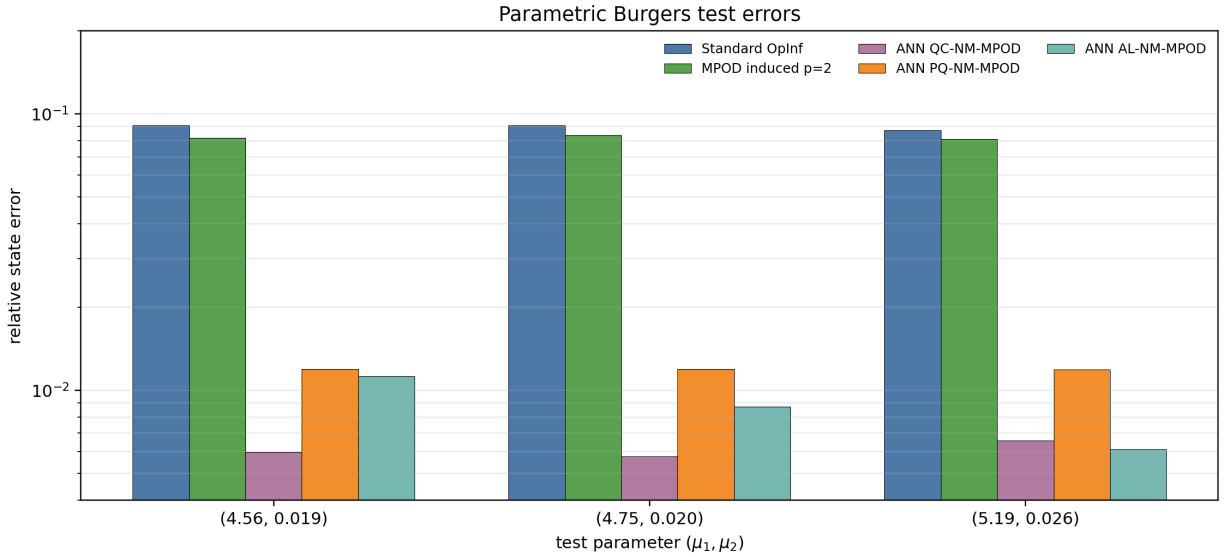


Figure 6: Parametric Burgers test errors. The ANN QC-NM-MPOD model gives the lowest average continuous-time error in the retained suite; the AL variant is cheaper and competitive, and is best at the third test point.

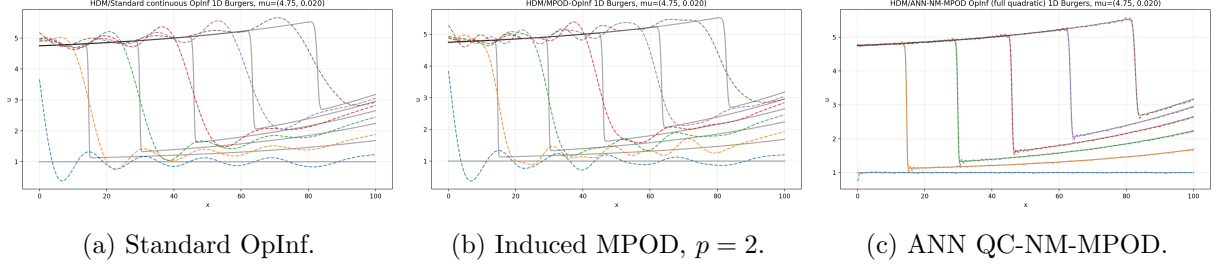


Figure 7: Representative Burgers rollout overlays at $\mu = (4.75, 0.020)$. The induced MPOD model improves strongly on the linear-subspace standard model, and the ANN QC-NM-MPOD decoder/operator pair gives the smallest continuous-time error in this comparison.

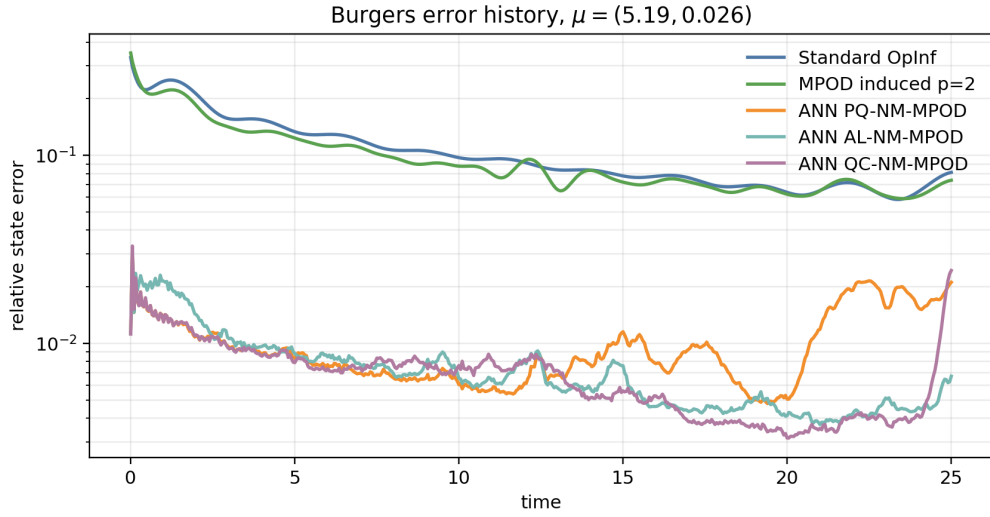


Figure 8: Error histories at $\mu = (5.19, 0.026)$ for the retained continuous-time Burgers comparison.